

CV Inconsistency for DAG Structure Learning

Findings on Corollary 6 of Lyu, Tai, Kolar, Aragam (AISTATS 2024)

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I replicated the CV inconsistency result from Corollary 6 of Section 5 in *Inconsistency of Cross-Validation for Structure Learning in Gaussian Graphical Models* in three settings: known ordering across varying graph sizes, unknown ordering, and non-Gaussian noise. All experiments use nodewise Lasso regression with the true ordering known unless stated otherwise, comparing CV, BIC, and EBIC.

Experiment 1: Known Ordering, Varying Graph Size

I simulated data from random Gaussian DAGs at $p = 5, 10, 20$ with known ordering, comparing CV, BIC, and EBIC across $n \in \{100, 200, 500, 1000, 2000, 5000\}$, averaged over 8 trials.

This confirms Corollary 6, as CV's SHD never reaches zero no matter how large n gets, while EBIC keeps improving. The gap gets worse at larger p : at $p = 20$, CV sits around 52–63 SHD while EBIC gets down to 27–39. Interestingly, CV's TPR is high, as it finds the real edges, but its FDR stays stubbornly high too, meaning it's also including fake edges, highlighting the inconsistency (CV doesn't miss real edges but it adds an abundant amount of fake ones). Moreover, EBIC did beat BIC in higher dimensions.

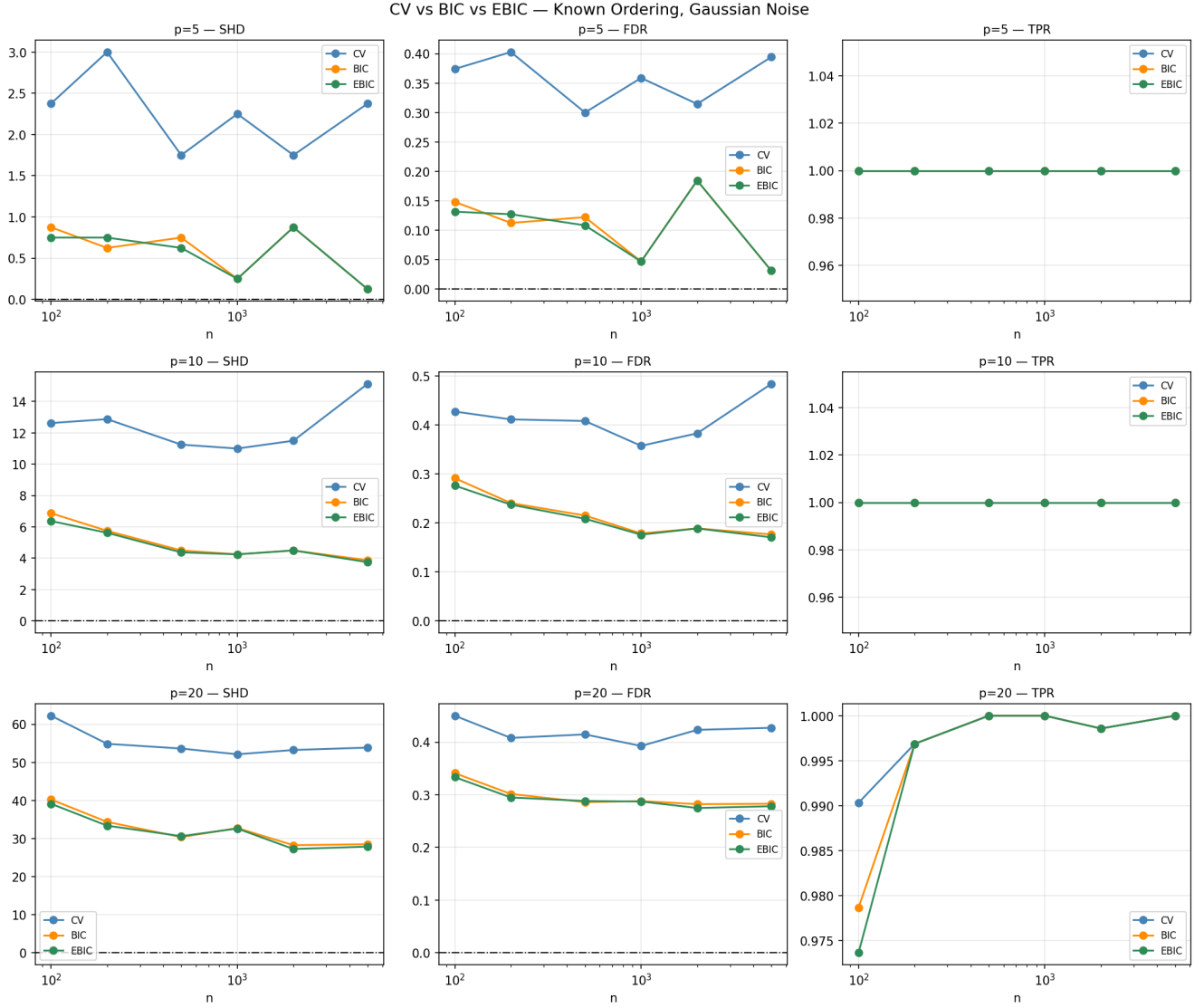


Figure 1: SHD, FDR, and TPR vs. sample size for CV, BIC, and EBIC across $p = 5, 10, 20$ (known ordering). The dashed line at zero indicates perfect recovery.

Experiment 2: Unknown Ordering

Corollary 6 assumes you know the true ordering, but in reality, you almost never do, so I was interested in researching more into this nuance, so I compared known vs. unknown ordering at $p = 5$, using a randomly permuted ordering as a stand-in for the unknown case.

Unknown ordering roughly doubles SHD across all the methods, as CV goes from ≈ 2.5 to ≈ 5.5 . This actually suggests that Corollary 6 is optimistic: it gives CV the easiest possible version of the problem, and CV still fails, as in practice where the ordering is unknown, things are strictly worse.

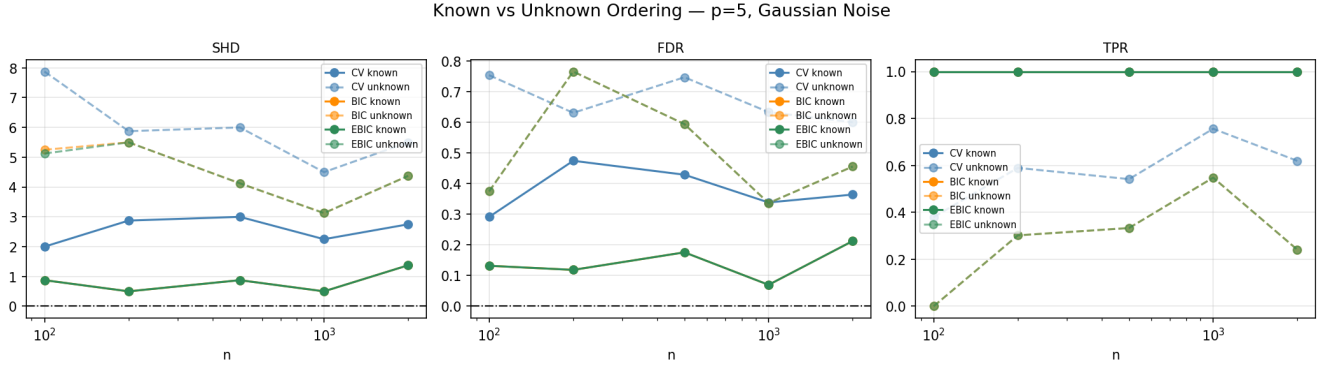


Figure 2: Known vs. unknown ordering for $p = 5$. Dashed lines indicate unknown ordering.

Experiment 3: Non-Gaussian Noise

Lastly the conclusion of the paper mentions extending the proof to non-Gaussian settings as an open problem, which I personally found very interesting due to its real-world applications (we never know if the noise is truly Gaussian or not). Thus, I tested CV on DAGs with Laplace and Uniform noise to see if the inconsistency result still holds true.

In fact, CV stays inconsistent across all three noise types, with SHD hovering around 1.5–3.0 and never going to zero. The failure doesn't seem to care much about the noise distribution, which suggests the inconsistency is driven by the prediction-vs-structure objective mismatch rather than anything specific to Gaussian assumptions.

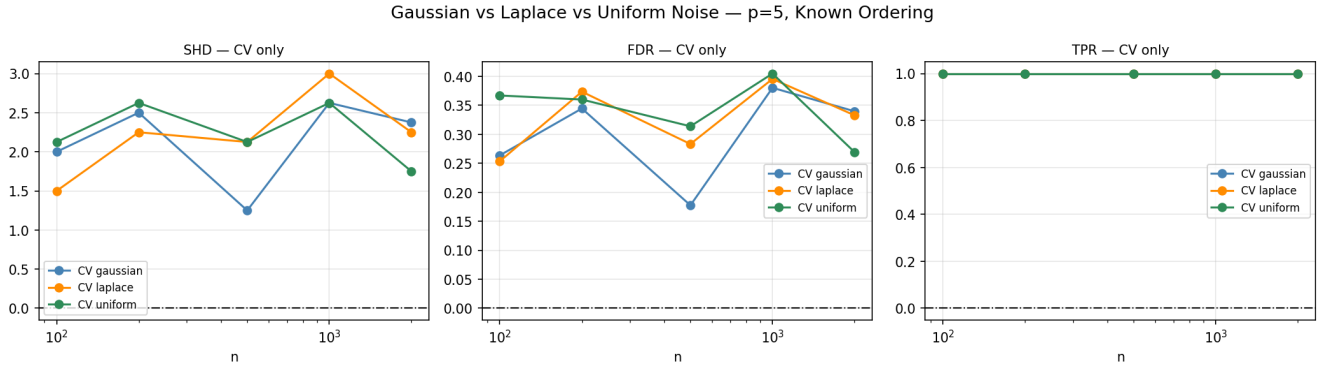


Figure 3: CV under Gaussian, Laplace, and Uniform noise ($p = 5$, known ordering).

Connection to CVCi and Broader Implications

The most fascinating result is the connection to my CVCi problem, as in that context, the CV also picks the lambda value to minimize prediction error on held-out experimental data, but what we actually want is the lambda that gives the most accurate ATE. Ultimately, those are different objectives for the same reason they are different in this context, as CV doesn't know the true difference between a real signal and a spurious one that happens to predict well on held-out data. I think this is worth investigating/researching further, and it makes me curious whether something like EBIC (which is shown to be better than CV in this problem) can be adapted/extended in the CVCi setting as well.